RK-4 (APPLICATION)

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K1: Acceleration at the start:

$K1_{ax} = 1.087 \times 10^{-6}\ Km/s^2$

$K1_{ay} = -6.035 \times 10^{-6}\ Km/s^2$

K2: Acceleration at ESTIMATED midpoint:

$x_{mid_1} = x_0 + V_{x_0} \frac{\Delta t}{2}$

$= -26\,072\,138.45 - 29.78893 \times 1800$

$= -26072138.45 - 53620.07$

$= -26\,125\,758.52\ Km$

$y_{mid_1} = y_0 + V_{y_0} \frac{\Delta t}{2}$

$= 144\,774\,673.82 - 5.39627 \times 1800$

$= 144774673.82 - 9713.29$

$= 144\,764\,960.53\ Km$

$r_{mid_1} = \sqrt{x_{mid_1}^2 + y_{mid_1}^2} \approx \sqrt{2.16395 \times 10^{16}}$

$\approx 147\,106\,000$

$\frac{Gm_{sun}}{r_{mid_1}^2} \approx 6.132 \times 10^{-6}\ Km/s^2$

$\frac{x_{mid_1}}{r_{mid_1}} \approx -0.177596$

$\frac{y_{mid_1}}{r_{mid_1}} \approx 0.984084$

Finally,

$K2_{ax} = -6.132 \times 10^{-6} \times (-0.177596)$

$\approx 1.089 \times 10^{-6}\ Km/sec^2$

$K2_{ay} = -6.132 \times 10^{-6} \times (0.984084)$

$\approx -6.036 \times 10^{-6}\ Km/sec^2$

K_3: Acceleration at refined midpoint

$V_{x_{mid}} = V_{x_0} + K1_{ax} \frac{\Delta t}{2}$

$= -29.78893 + 1.087 \times 10^{-6} \times 1800$

$= -29.78893 + 0.001957$

$= -29.78697$

$V_{y_{mid}} = V_{y_0} + K1_{ay} \frac{\Delta t}{2}$

$= -5.39627 + -6.035 \times 10^{-6} \times 1800$

$= -5.39627 - 0.010863$

$= 5.40713\ Km/s$

$x_{mid_2} = x_0 + V_{x_{mid}} \frac{\Delta t}{2}$

$= -26072138.45 - 29.78697 \times 1800$

$= -26\,125755\ Km$

$y_{mid_2} = y_0 + V_{y_{mid}} \frac{\Delta t}{2}$

$= 144774673.82 - 5.40713 \times 1800$

$= 144764940.98\ Km$

$r_{mid_2} = \sqrt{x_{mid_2}^2 + y_{mid_2}^2} \approx 147\,106\,000\ Km$

$K3_{ax} = -6.132 \times 10^{-6}\ (-0.177596)$

$\approx 1.0891 \times 10^{-6}\ Km/s^2$

$K3_{ay} = -6.132 \times 10^{-6}\ (0.984083)$

$\approx 6.0350 \times 10^{-6}\ Km/s^2$

K4: Endpoint acceleration

$x_{end} = -26\,179364.49\ Km$

$y_{end} = 144\,755\,169.02\ Km$

$r_{end} = \sqrt{x_{end}^2 + y_{end}^2}$

$= \sqrt{6.8535 \times 10^{14} + 2.0954 \times 10^{16}}$

$= \sqrt{2.1639 \times 10^{16}}$

$\approx 147\,106\,000$

$\frac{Gm_{sun}}{r_{end}^2} \approx 6.132 \times 10^{-6}\ Km/s^2$

$\frac{x_{end}}{r_{end}} = \frac{-26\,179364.49}{147\,106\,000}$

≈ -0.177991

$\frac{y_{end}}{r_{end}} = \frac{144\,755\,169.02}{147\,106000}$

≈ 0.983814

$K4_{ax} = -6.132 \times 10^{-6} (-0.177991)$

$\approx 1.0913 \times 10^{-6}\ Km/s^2$

$K4_{ay} = -6.132 \times 10^{-6} \times 0.983184$

$\approx -6.0334 \times 10^{-6}\ Km/s^2$

$K1_{ax} \approx 1.0870 \times 10^{-6}$

$K1_{ay} \approx -6.0350 \times 10^{-6}$

$K2_{ax} \approx 1.0890 \times 10^{-6}$

$K2_{ay} \approx -6.0360 \times 10^{-6}$

$K3_{ax} \approx 1.0891 \times 10^{-6}$

$K3_{ay} \approx -6.0350 \times 10^{-6}$

$K4_{ax} \approx 1.0913 \times 10^{-6}$

$K4_{ay} \approx -6.0334 \times 10^{-6}$